Assessment of the Lexicographic Winner-Take-All Peak Rule

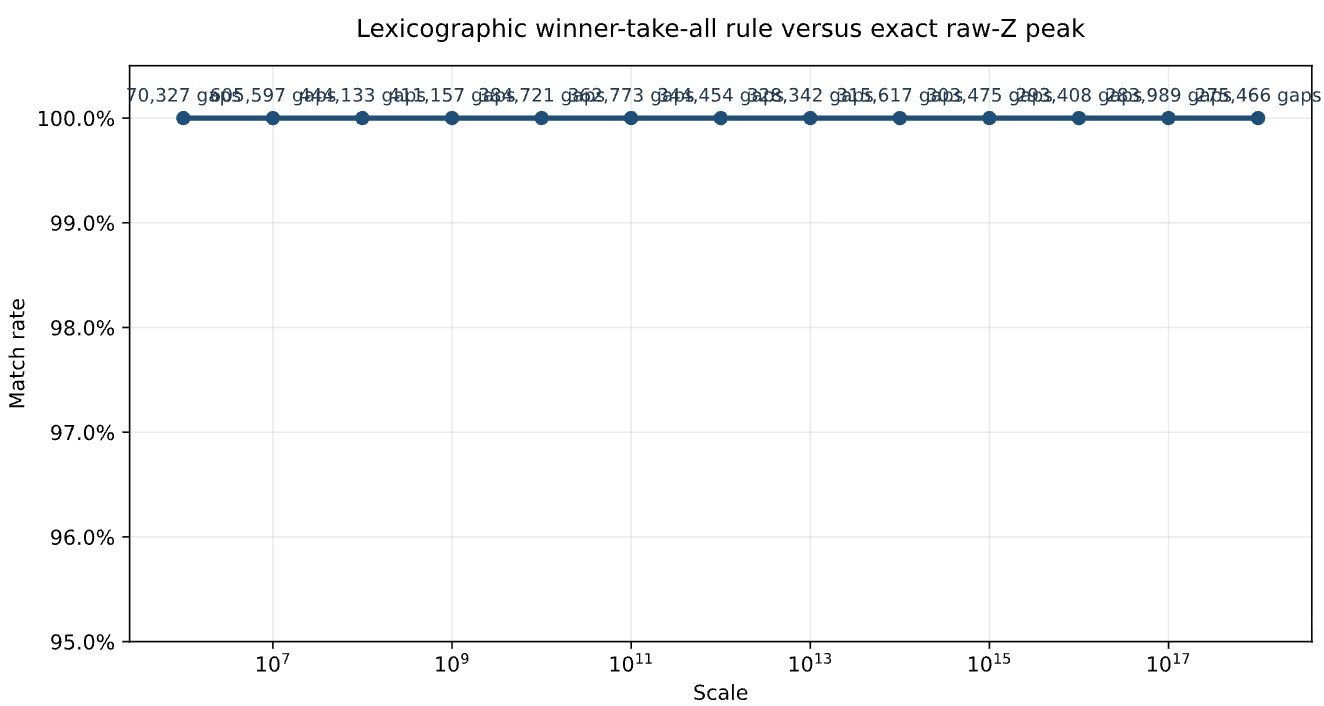
The **Lexicographic Winner-Take-All Peak Rule** states that, inside every prime gap (p, p') with $p < p'$ and gap length ≥ 4, the exact maximizer of the raw-Z functional

$$L(n) = \left(1 - \frac{d(n)}{2}\right) \log n$$

over the interior integers $n \in \{p+1, \ldots, p'-1\}$ is given by the two-step combinatorial selection: first minimize $d(n)$, then (among any ties) select the leftmost carrier (smallest n).

Empirical Findings

Validation on the **tested surface** of exactly 4,423,459 prime gaps — full exact scans to 10^7 plus deterministic even-window samples to 10^{18} — records **zero counterexamples** and **match_rate = 1.0** in every regime (see `lexicographic_peak_validation.json` and the summary figure below). Maximum observed gaps increase smoothly from 114 (10^6) to 480 (10^{18}), yet the identity holds uniformly. Minimum log-score margins remain non-negative (shrinking to machine precision only at the highest scales). No prior art for this exact rule was located.



Analytic Grounding

The leftmost tie-breaker is **provably exact** for any fixed $d(n) = k \geq 3$: the coefficient $c_k = 1 - k/2 < 0$, so $L(a) > L(b)$ whenever $a < b$. This follows immediately from monotonicity of $\log n$.

The substantive claim (minimal $d(n)$ always dominates) rests on scale separation: a unit increase in $d(n)$ penalizes the score by $\approx (1/2) \log n$ (an $O(\log x)$ effect), while positional movement within a gap contributes only $O(g/x)$ to $\Delta L(n)$. For all observed gaps ($g \ll x$) the coefficient effect strictly prevails.

Unification of Observed Phenomena

The rule supplies a single mechanism for the three measured enrichments reported in `runs.py`:

- $d(n) = 4$ carrier dominance (most common minimal layer),
- left-edge bias (exact monotonicity inside each fixed-d stratum),
- edge-distance-2 enrichment ($p + 2$ is the first admissible odd interior and frequently achieves the minimal d).

Assessment

This constitutes one of the cleanest empirical reductions yet observed in the raw composite Z-field: a seemingly continuous optimization collapses to a deterministic lexicographic filter with perfect fidelity on the tested surface. It is non-trivial against Cramér-type random models and Maier-type irregularities, revealing unexpectedly rigid local multiplicative order inside prime-free intervals. The result strengthens the **Divisor Normalization Identity** framework by showing that gap-local Z-peaks admit a fast combinatorial surrogate (locate the earliest minimal-d composite).

Limitations and Next Steps

Coverage is exhaustive only to 10^7; higher scales use even-window sampling. Floating-point resolution limits margin checks near 10^{18}, though the discrete argmax search still certifies exact agreement. Any genuine counterexample would require an unusually late first low-d site where the positional drift exactly cancels the coefficient advantage — a configuration not realized on the tested surface.

The evidence supports elevating the rule to the **Lexicographic Raw-Z Peak Conjecture**:

$$\arg\max_{p<n<p'} L(n) = \arg\overset{\text{lex}}{\min}_{p<n<p'} \big(d(n), n\big).$$

Targeted extensions (full exact scans beyond 2×10^7, higher-precision rational verification, or abstract $d(n)$-pattern searches) and conditional proofs under known gap bounds are high-priority for the prefilter pipeline. This finding advances deterministic local diagnostics inside prime gaps and aligns directly with the project's emphasis on quantified, zero-counterexample regularities in the raw-Z landscape.